

Methods and data sources

Everything describing the county is real; every assessment answer is synthetic. This note sets out where each number comes from, how hazard exposure is measured, how catchments and scores are built, and what the figures cannot be used for. Numbers in blue refer to the reference list on page 2.

WHAT IS REAL HERE AND WHAT IS NOT

Everything describing the county is real. The facility register, MFL codes, KEPH levels, sub-counties and wards come from the Ministry's own registry [2](#). Facility coordinates are surveyed positions [3](#). Rainfall, temperature, heat days, extreme rainfall and wind gusts are ERA5 reanalysis [5](#); river discharge is GloFAS [6](#); wealth is the Relative Wealth Index [7](#); population is the 2019 census [8](#); boundaries are OpenStreetMap [9](#); travel-time access is published work [4](#).

Every assessment answer is synthetic. Readiness and vulnerability scores, staffing figures, water and power readings, narrative gaps and recommended actions were generated to exercise the tool ahead of fieldwork. They are not collected data and must not be read as findings about any named facility. Replace the derived-scores file with a real REDCap export [1](#) and everything else on this page works unchanged.

WHICH FACILITIES ARE INCLUDED

The register is the government-owned (Ministry of Health) subset of the Kenya Master Health Facility Registry for Kilifi: 148 facilities [2](#). The registry's public export also lists faith-based and NGO facilities, which Kenyan practice counts as public sector; those are excluded here so that everything on this page refers to facilities the county runs.

41 of the 148 carry an assessment in this build. They were selected by the county. Facilities are matched to the registry on MFL code rather than name, because the name a county uses and the name in the registry often differ — Rabai Sub County Hospital is registered as Rabai Rural Health Demonstration Centre, and Marafa Sub County Hospital as Marafa Health Centre.

WHERE THE FACILITIES ARE ON THE MAP

Coordinates come from the sub-Saharan Africa public health facility database [3](#), joined to the registry by facility name with manual review of every uncertain match. 99 of the 148 have a position, 93 of them GPS-surveyed. The 49 without one were registered after that database's February 2019 snapshot and are almost all Level 2 dispensaries, concentrated in Ganze, Kilifi North, Rabai and Magarini.

Four assessed facilities fall in that gap — Jila Model Health Centre, Mrima Wa Ndege Dispensary, Shakahola Dispensary and Chilodi CrossFit Health Centre. They are characterised at their sub-county centroid so they can still be scored, are flagged wherever they appear, and are left off the map rather than given a pin that implies a precision we do not have.

Maps use an equirectangular projection with a $\cos(\text{latitude})$ correction, so a kilometre east is the same length on screen as a kilometre north. Each map carries a 20 km scale bar.

HOW HAZARD EXPOSURE IS MEASURED

Five of the six ECHRRA-K hazards are measured at each facility's true position rather than reported by the facility, then banded onto the tool's 0–5 exposure scale. **Extreme temperature** takes the higher of two bands, days above 35 °C and mean daily maximum, because the humid coastal strip rarely passes 35 °C dry-bulb but still carries real heat stress [5](#). **Drought** is banded on mean annual rainfall, inverted [5](#). **Heavy rainfall and flooding** is banded on Rx5day and raised one band where a GloFAS cell within 20 km carries a mean annual maximum discharge above 50 m³/s [5](#) [6](#). **Strong winds** is banded on mean annual maximum 10 m gust against an absolute scale, not the county's own spread: Kilifi runs 59–70 km/h throughout, Beaufort 7–8, so a county-relative band would invent differences that are not there [5](#). **Sea-level rise** is banded on distance to the coastline and reaches 20 km inland [9](#).

Landslides are not derived from data. There is no terrain layer in this build, so landslide exposure is assigned to the Rabai–Kaloleni hill ridge and to nowhere else. It should be replaced with a slope and soil surface before the figure is used.

ECHRRA-K HAZARD	MEASURED QUANTITY	BANDING NOTE	REF
Extreme temperature	days/yr > 35 °C, and mean daily max °C	higher of the two bands	5
Drought	mean annual rainfall, mm	inverted: drier is higher	5
Heavy rainfall / flooding	Rx5day, mm; river discharge, m³/s	+1 band above 50 m³/s within 20 km	5 , 6
Strong winds	mean annual max 10 m gust, km/h	absolute scale, Beaufort-aligned	5
Sea-level rise	distance to coastline, km	0 beyond 20 km inland	9
Landslides / mudslides	— not derived from data	assigned to the Rabai–Kaloleni ridge	—

WHO LIVES WHERE, AND WHICH FACILITY THEY USE

The population view changes the unit of analysis from the facility to the populated place. Each of the 1,035 Relative Wealth Index cells inside the county [7](#) is assigned to a sub-county by point-in-polygon against the OpenStreetMap boundaries [9](#), and that sub-county's 2019 census population [8](#) is spread evenly across its cells. This is a distributed total, not a modelled population surface: treat the counts as indicative of scale.

Cells are then split into five equal population groups by wealth index and attached to the nearest facility by straight-line distance, and separately to the nearest KEPH Level 4 facility, because a dispensary cannot take a caesarean or a severe-malaria admission. A cell counts as heat-exposed above 20 days a year over 35 °C, and as flood-exposed within 20 km of a modelled river carrying more than 50 m³/s at its annual peak. Extreme rainfall is deliberately not used as the flood threshold: in Kilifi the wettest ground is the wealthiest coastal strip, so a rainfall cut-off would rank the tourist coast as the priority.

Straight-line distance ignores roads, rivers and terrain. The honest measure of access is the published travel-time modelling [4](#), reported alongside at sub-county level; it is not available per facility, so Euclidean distance stands in for the catchments.

HOW THE SCORES ARE BUILT

Overall risk is $\sqrt{(\text{exposure} \times \text{severity}) \div 5 \times (\text{weighted weakness} \div 5)} \times 100$, on a 0–100 scale. Exposure and severity are the means across the six hazards; weakness is the criticality-weighted mean across the eight summary systems in ECHRRRA-K section K [1](#). Per-hazard risk uses the same geometric form so a band label means the same thing everywhere.

The ECHRRRA-K rubrics run in two directions, so red, amber and green are assigned by meaning and never by the raw number. Tables 3 and 7–12 (utilities, workforce, supplies, continuity, alternatives, leadership, early warning) run 1 worst to 5 best; tables 1, 6 and 13 (exposure, severity, vulnerability) run 1 best to 5 worst.

On overall risk, **red is 35 and above, amber 25 to 34, green below 25**. Red is a site carrying real hazard whose critical systems score around 4 of 5 for weakness. Most facilities land in amber, because the composite averages ten weighted systems and averages cluster; the traffic light flags the tails, and the per-system chips are where the detail is.

KNOWN LIMITATIONS

- Assessment answers are synthetic. Any apparent relationship between how poor a catchment is and how weak its facility scores is an artefact of the generator, not a finding.
- Catchment distance is straight-line, not travel time. The published travel-time figures are reported alongside but exist only at sub-county resolution.
- 49 of the 148 registered facilities have no coordinate, so catchments are drawn against the 99 that do; real distances are somewhat shorter than shown.
- Population is a sub-county census total spread evenly over populated cells, not a gridded population surface. Treat counts as indicative of scale.
- Landslide exposure is not derived from data; there is no terrain layer in this build.
- Sub-county boundaries are simplified to 150 vertices — suitable for mapping, not for area calculations.

REFERENCES

- 1 ECHRRRA-K v3.0** — Ministry of Health, Republic of Kenya (2026). Extreme Weather and Climatic Events Health Facility Risk and Readiness Assessment tool (ECHRRRA-K), version 3.0, January 2026.
The assessment instrument itself: the 13 scoring rubrics, the six hazards, the ten facility systems and the KEPH level definitions. Responses are held in REDCap; the export here uses the live 690-column layout.
- 2 KMHFR** — Ministry of Health, Republic of Kenya. Kenya Master Health Facility Registry. Public facility export, September 2026. <https://kmhfr.health.go.ke/>
The facility register: official name, MFL code, KEPH level, facility type, sub-county, ward, ownership and bed count. This tool uses the government-owned (Ministry of Health) subset only — 148 facilities in Kilifi. The registry's public export also contains faith-based and NGO facilities; those are excluded here.
- 3 Maina et al. 2019** — Maina J, Ouma PO, Macharia PM, Alegana VA, Mitto B, Fall IS, Noor AM, Snow RW, Okiro EA (2019). A spatial database of health facilities managed by the public health sector in sub-Saharan Africa. *Scientific Data* 6:134. <https://doi.org/10.1038/s41597-019-0142-2>
Facility coordinates, joined to the registry by facility name. 99 of the 148 government facilities in Kilifi carry a position; 93 of those were surveyed by GPS and the rest digitised. The remaining 49 were registered after this February 2019 snapshot and have no coordinate.
- 4 Moturi et al. 2022** — Moturi AK, Suiyanka L, Mumo E, Snow RW, Okiro EA, Macharia PM (2022). Geographic accessibility to public and private health facilities in Kenya in 2021: an updated geocoded inventory and spatial analysis. *Frontiers in Public Health* 10:1002975. <https://doi.org/10.3389/fpubh.2022.1002975>
Published travel-time accessibility by sub-county: the share of the population within one and two hours of a health facility. Modelled in AccessMod 5.6 by cost-distance at 300 m over road network, land cover and elevation surfaces, for walking, cycling and motorised transport. Population projected to 2021.
- 5 ERA5** — Hersbach H, Bell B, Berrisford P, et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146:1999–2049. Retrieved through the Open-Meteo Historical Weather API. <https://doi.org/10.1002/qj.3803>
Climate at each facility and each populated cell, on a 72-point grid across the county, 2005–2024: mean annual precipitation, mean daily maximum temperature, days per year above 35 °C, Rx5day (mean annual maximum consecutive five-day rainfall) and mean annual maximum 10 m wind gust.
- 6 GloFAS** — Copernicus Emergency Management Service. Global Flood Awareness System (GloFAS) river discharge reanalysis. Retrieved through the Open-Meteo Flood API. <https://global-flood.emergency.copernicus.eu/>
Riverine flood exposure: mean annual maximum daily river discharge, 2005–2024, m³/s, on the GloFAS model grid. Shown at the model's own cells rather than interpolated, because discharge follows rivers and interpolation between cells invents water where there is none.
- 7 Relative Wealth Index** — Chi G, Fang H, Chatterjee S, Blumenstock JE (2022). Microestimates of wealth for all low- and middle-income countries. *PNAS* 119(3):e2113658119. Meta Data for Good Relative Wealth Index, retrieved via the Humanitarian Data Exchange. <https://doi.org/10.1073/pnas.2113658119>
Relative wealth on a ~2.4 km grid, used to split the county's population into five equal groups. The index is relative, not a poverty rate: 0 is the national mean and negative values are poorer. 1,035 of its cells fall inside Kilifi.
- 8 KNBS Census 2019** — Kenya National Bureau of Statistics (2019). Kenya Population and Housing Census, Volume I: Population by County and Sub-County. Sub-national totals retrieved via the Humanitarian Data Exchange (cod-ps-ken). <https://www.knbs.or.ke/>
Population by sub-county, including female, male and under-five counts. County total 1,453,762.
- 9 OpenStreetMap** — OpenStreetMap contributors. Kilifi County relation 3495545 and the seven admin_level=6 sub-county relations, retrieved through the Overpass API. Licensed under the Open Database License (ODbL). <https://www.openstreetmap.org/relation/3495545>
The county outline and the seven sub-county boundaries drawn on every map. Member ways stitched into closed rings and simplified with Douglas-Peucker to 150 vertices each.